

Ibrahim Abiodun Imam

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Summary:

I am a Ph.D. candidate in Chemical Engineering at the University of Kentucky, specializing in computational chemistry for molecular discovery. My research integrates molecular modeling, deep learning, and quantum chemical calculations for mechanistic study and property predictions. With expertise in molecular modeling, machine learning, and high-performance computing, I have developed pipelines for high throughput screening, and molecular property validation. My work has resulted in peer-reviewed publications and presentations at leading conferences.

Education:

Ph.D. in Chemical Engineering

University of Kentucky, Lexington, KY

August 2022 – December 2026 (Expected)

GPA: 3.92

Skills:

- **Machine Learning:** Data science, traditional ML (Scikit-learn, XGBoost), deep learning (Pytorch).
- **Cheminformatics:** RDkit, Obabel, scaffold hopping and similarity search.
- **Molecular Modeling & Simulations:** GROMACS, OpenMM, ORCA, AutoDock Vina, Pymol, Avogadro.
- **Free Energy Calculations:** Alchemical methods (FEP, TI, MBAR), MM/GBSA.
- **Enhanced Sampling:** Well-tempered metadynamics, umbrella sampling (PLUMED).
- **Programming & Scripting:** Python, R, Linux bash, Git.
- **Process Design:** Material/energy balance, unit operations, reactor design, process modelling.

Experience:

Graduate Research Assistant

University of Kentucky, Lexington, KY

Aug 2022 – present

- Developed a multi-scale modeling pipeline integrating MD simulations and deep learning (PyTorch) for molecular discovery (small molecule and peptides), **resulting in 2 peer-reviewed publications**.
- Curated and preprocessed small molecule and protein datasets for molecular docking, MD simulations, and machine learning model training, **improving prediction accuracy by 10%**.
- Performed unbiased MD simulations and enhanced sampling to investigate mutation-induced drug resistance and functional loss in proteins, **speeding up mechanistic insight for drug discovery**.
- Automated high-performance computing (HPC) workflows using Python and Bash scripts, **reducing simulation setup time by 45%**.
- Collaborated with interdisciplinary teams to integrate computational and experimental approaches for drug repurposing, **contributing to the discovery effort for EGFR positive cancer**.

Graduate Research Fellow

Lawrence Berkeley National Laboratory, Berkeley, CA

June 2023 – Aug 2023

- Enhanced NWChemEx for biochemical system modeling, improving scalability.
- Automated high-performance computing (HPC) workflows using python and bash scripts.
- Conducted benchmarking analyses to validate software enhancements.

Publications:

1. Ibrahim A. Imam, *et al.* "L858R/L718Q and L858R/L792H Mutations of EGFR Inducing Resistance Against Osimertinib." *Proteins: Structure, Functions, and Bioinformatics*, 2025.
2. Ibrahim A. Imam, *et al.* "Integrating Protein Language Model and Molecular Dynamics Simulations to Discover Anti-Biofouling Peptides." *Langmuir*, 2024.

Projects:

- **Discovery of Anti-Biofouling Peptides:** Integrated protein language models and MD simulations to identify peptides from microbiomes, resulting in a publication in *Langmuir* (2024).
- **EGFR Mutation Resistance Study:** Investigated L858R/L718Q and L858R/L792H mutations using MD simulations and quantum chemical calculations, uncovering resistance mechanisms against osimertinib. Published in *Proteins: Structure, Functions, and Bioinformatics* (2025).
- **Small Molecule Screening Pipeline:** Developed an in-silico pipeline for identifying small molecules to overcome EGFR mutation-induced drug resistance, presented at AIChE Annual Meeting (2023).

Workshops:

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| ▪ Virtual School on Machine Learning for Molecules
Scope: Data cleaning/imputation Molecular featurization ML & DL
<i>NSF - 9th i-CoMSE workshop</i> | April 2025 |
| ▪ MDAnalysis/MolSSI Workshop at Arizona State University
Scope: Software development Software version control
<i>MDAnalysis and Molecular Sciences Software Institute (MolSSI)</i> | June 2024 |
| ▪ Methods for Enhanced Sampling (Virtual)
Scope: Umbrella sampling, Metadynamics Replica exchange
<i>NSF - 3rd i-CoMSE Workshop</i> | April 2023 |